

Anirudh Sai Nagula

Data Engineer

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PROFESSIONAL SUMMARY :

- Data Engineer with 6+ years of experience in designing, building, and optimizing ETL/ELT pipelines, big data processing, cloud computing, and data warehouse applications.
- Expertise in AWS, GCP, and Snowflake, leveraging Data Vault 2.0 methodologies for scalable and flexible data modeling.
- Proficient in Python, PySpark, SQL, and Java, implementing enterprise-grade data transformation using Apache Spark, Airflow, and Hadoop.
- Strong background in workflow orchestration, metadata management, data governance, and compliance frameworks ensuring data integrity and regulatory adherence.
- Experienced in CI/CD pipeline integration, DevOps best practices, and infrastructure automation to improve data engineering efficiency.
- Proven ability to collaborate with cross-functional teams to drive business intelligence and analytical initiatives.

TECHNICAL SKILLS :

Databases & Warehousing: Snowflake, AWS RDS, MySQL, Microsoft SQL Server, PostgreSQL, Palantir Foundry, MongoDB, Amazon Redshift, Hive.

Programming & Scripting: Python, SQL, Java, C++, JavaScript, Boto3, Shell Scripting.

Cloud & DevOps: AWS (S3, EC2, Lambda, Redshift, Glue, Step Functions, CloudFormation, DMS, EMR), GCP (Dataflow, Cloud Functions, Dataproc, Pub/Sub, Composer), Docker, Kubernetes, Jenkins, Terraform.

Big Data & ETL Processing: Apache Spark, PySpark, Apache Airflow, AWS Glue, Apache Hudi, Sqoop, Kafka.

Workflow Orchestration & Automation: Apache Airflow, Google Cloud Composer, AWS Step Functions.

Data Modeling & Governance: Data Vault 2.0, Erwin, Data Quality Frameworks, Master Data Management (MDM), Metadata Management.

Visualization & Reporting: Tableau, Power BI, D3.js, Chart.js.

PROFESSIONAL EXPERIENCE :

Humana, Louisville, KY | Data Engineer |

Dec 2023 – Present

- Architected and implemented a scalable data lake solution on Snowflake, handling 350+ GB of daily data ingestion and processing while ensuring optimized query performance and cost efficiency.
- Designed and developed automated ETL workflows using Apache Airflow and PySpark, integrating multiple structured and semi-structured data sources from JSON, Parquet, CSV, and XLSX formats.
- Implemented dynamic data masking and encryption strategies to ensure PII data compliance across all data repositories, adhering to industry security and governance standards.
- Developed Snowflake SQL stored procedures and performance-tuned queries to enhance data accessibility and reduce processing times for business-critical analytics workloads.
- Provisioned AWS infrastructure resources, including EC2, S3, IAM roles, and bucket policies, ensuring a secure and scalable cloud environment for data operations.

Environment: AWS (S3, EC2, Lambda, RDS, CloudFormation, Step Functions), Snowflake, Palantir Foundry, Python, PySpark, Apache Airflow, DBT Vault, Data Vault 2.0, GitHub, Jenkins, Tableau.

Fannie Mae, Plano, TX | Data Engineer |

Mar 2023 – Nov 2023

- Developed and optimized end-to-end data pipelines for the Customer Data Platform (CDP), ensuring seamless data integration and synchronization across multiple internal systems.
- Designed and deployed AWS-native solutions incorporating Apache Hudi for efficient upsert operations and SCD-2 implementation, enhancing data historization and versioning strategies.
- Built an ETL pipeline from AWS S3 through various data transformation stages to Redshift, leveraging AWS Glue and PySpark for data cleansing and enrichment.

- Automated data migration from legacy Oracle and Hadoop systems to Redshift using AWS DMS, ensuring minimal downtime and data integrity.
- Implemented monitoring and alerting mechanisms for data processing failures using AWS CloudWatch, SNS, and PagerDuty, improving system reliability and responsiveness.

Environment: AWS (S3, Redshift, Lambda, Glue, DMS, EMR, EC2, Step Functions), Apache Hudi, PySpark, Airflow, Snowflake, Git, Jenkins, Power BI.

DXC Technologies, India | Data Engineer |

Apr 2021 – Jun 2022

- Designed and developed high-performance ETL pipelines in Hive and Python to extract, transform, and load large-scale datasets from multiple sources, optimizing processing speed and efficiency.
- Built and orchestrated real-time data streaming solutions using GCP Pub/Sub and Dataflow, enabling near-instant data processing and analytics.
- Developed and implemented SQL and PL/SQL scripts for complex data transformation logic in BigQuery, significantly improving query performance and resource utilization.
- Automated and managed data ingestion from Oracle to GCP storage layers using Apache Sqoop and custom Python scripts, ensuring data consistency across hybrid environments.
- Created robust workflow orchestration pipelines using Google Cloud Composer, automating task dependencies and reducing manual intervention in data processing workflows.

Environment: GCP (Cloud Functions, Pub/Sub, Dataflow, Dataproc, BigQuery, Cloud Composer), Hive, Apache Sqoop, Python, SQL, Git, Jenkins.

DBS Bank, India | Data Engineer |

Jan 2019 – Mar 2021

- Led data engineering efforts in building a new digital banking platform, working closely with stakeholders to gather and translate business requirements into technical solutions.
- Developed core banking functionalities leveraging Java and Spring Boot, integrating with various payment gateways, credit bureaus, and regulatory APIs to enhance platform capabilities.
- Designed and optimized SQL stored procedures and complex query logic to support transactional data processing and financial reporting.
- Built interactive data visualization dashboards using D3.js, Chart.js, and Power BI, enabling real-time analytics and reporting for business users.
- Managed CI/CD pipelines using Azure DevOps, ensuring seamless deployments and version control for ongoing application enhancements and security updates.

Environment: Java, Spring Boot, React, Microsoft Azure, SQL Server, PL/SQL, D3.js, Chart.js, Git, Azure DevOps, Power BI.

EDUCATION :

Masters in Computer Engineering, University of North Texas, TX – 2024

Bachelors in Computer Science, JNTUH, India – 2019

CERTIFICATION :

[Microsoft Azure Data Engineer Associate \(DP-203\)](#)